

Trainings ▾

General Information

Note : There are two different piston and liner combinations available for this engine platform. The ferrous cast ductile (FCD) piston must be used in combination with a nitrided cylinder liner. The steel piston must be used in combination with a high boron liner. It is important that these parts are used in this combination stated and **never** mixed. An engine must **never** be run with a mixture of FCD and steel pistons. It must contain **only** a full set of FCD pistons or a full set of steel pistons. The liners can be distinguished from each other by appearance. The nitride liners will have a matte appearance on the outer circumference. The high boron liners are bright and shiny in appearance.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

To reduce the possibility of personal injury, avoid contact of hot oil with your skin.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/56/56-002-004-tr.html)
- Drain the engine lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)
- Remove the lubricating oil pan adapter. Refer to Procedure 007-027 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)
- Remove the lubricating oil pan suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)
- Remove the lubricating oil transfer tube. Refer to Procedure 007-040 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-040-tr.html)
- Remove the lubricating oil pump. Refer to Procedure 007-031 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)
- Remove and discard the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-089-tr.html)
- Remove the piston cooling nozzles for the cylinders requiring removal. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-046-tr.html)

Note : The block stiffener plate use has been discontinued for the QSK45 and QSK60 engines.

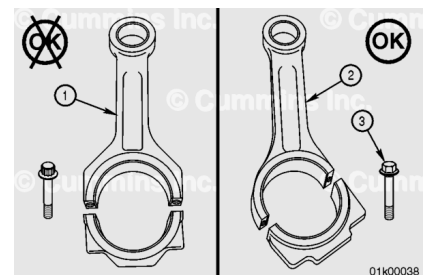
Remove

Use the following steps for engines containing angle split connecting rods (2) **only**.

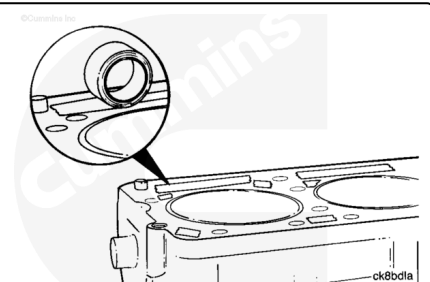
For engines containing straight split connecting rods (1). Refer to Procedure 001-045 in Section 1.

(/qs3/pubsys2/xml/en/procedures/56/56-001-045-tr.html)

Note : The angle split connecting rod (2) can be identified by 6 point head capscrews (3) and the angled joint.

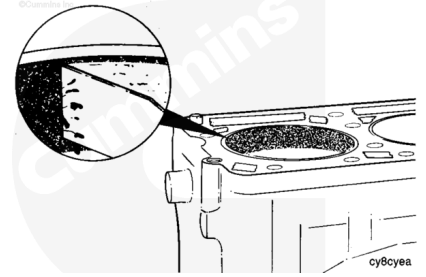


Use tape to protect the push rod galleys, coolant passages, and oil passages from contamination.



⚠ CAUTION ⚠

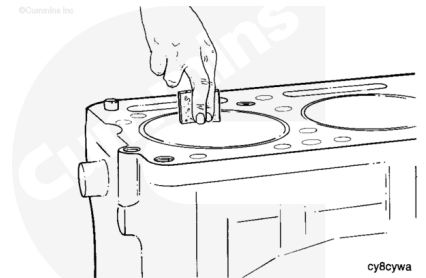
Do not use abrasive paper to remove the carbon deposits. Small particles of abrasive paper will cause severe engine damage.



Use a scraper or a similar blunt-edged tool to loosen the carbon deposits.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's instructions for use. Wear goggles and protective clothing to avoid personal injury.



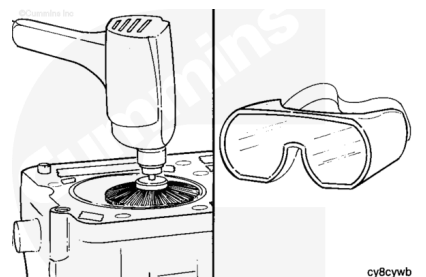
⚠ CAUTION ⚠

Make sure that no abrasive cleaners or materials are used in the piston ring travel area.

Remove the remaining carbon with an abrasive nylon pad, such as Scotch-Brite™ 7448, Part Number 3823258, and solvent. The carbon **must** be removed, but the surface does **not** have to look like new metal.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using a wire brush. Be sure the wire brush is rated for the correct rpm if the brush is motor driven. Flying debris and dirt can cause personal injury.

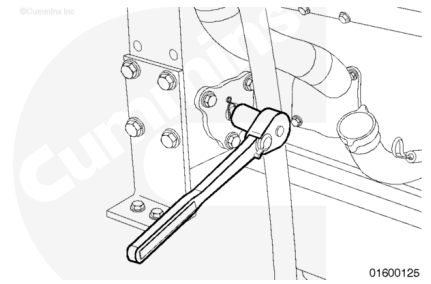


An alternative method to remove the carbon ridge is to use a high-quality steel wire wheel installed in a drill.

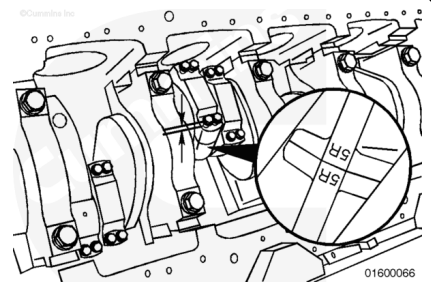
Do **not** use a steel wire wheel of inferior quality. It will lose steel bristles during operation and cause additional contamination.

Do **not** use a steel wire wheel in the piston ring travel area. Operate the wire wheel in a circular motion to remove the deposits.

Use the barring mechanism to rotate the engine. Rotate the crankshaft to position the connecting rod at bottom dead center.

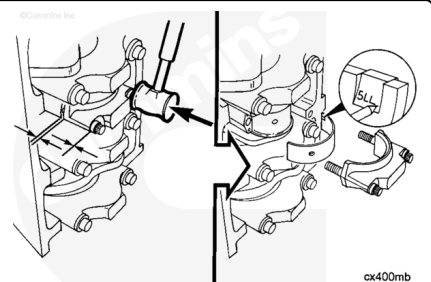


The connecting rods **must** have the cylinder number and bank marked on both the rod and the cap on the side positioned toward the camshaft. Check the rods for correct markings. Use a steel stamp to mark any connecting rod that is **not** correctly marked.



Loosen the capscrews until there is 6 mm [1/4 in] of clearance between the rod cap and the capscrew head.

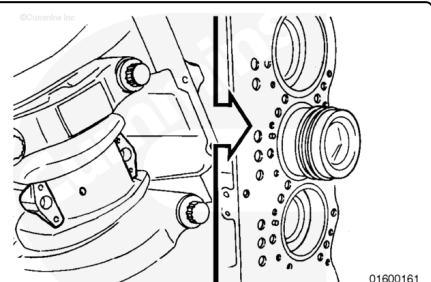
Use a mallet to tap the connecting rod capscrews until the connecting rod cap and rod separate. Remove the capscrews and the cap.



Remove the lower rod bearing. Use an awl to mark the bearing position in the tang area.

Mark the bearing for future identification or analysis.

Push the piston and connecting rod up until the piston rings are above the cylinder liner.



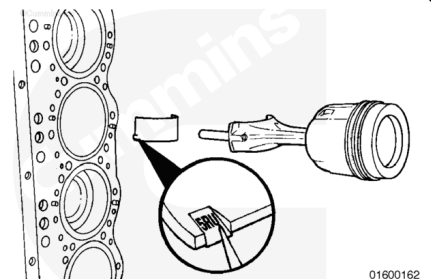
⚠ CAUTION ⚠

Put the piston and rod assembly in a rack to reduce the possibility of damage to the piston and rod assembly.

Remove the piston and connecting rod assembly.

Remove the upper connecting rod bearing.

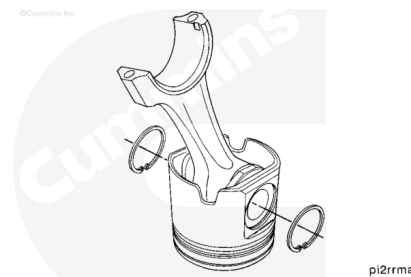
Use an awl to mark the bearing position in the tang area.



Disassemble

Use internal snap ring pliers to remove the snap rings from both sides of the piston.

Mark or tag the rings from each cylinder with the cylinder number and bank for analysis, if desired.



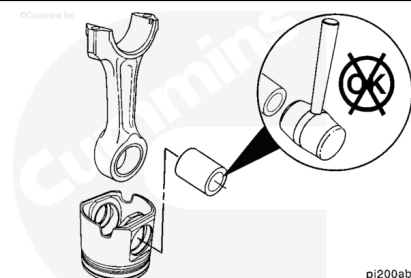
⚠ CAUTION ⚠

To prevent damage to the piston, do not drop the piston while removing the piston pin.

Remove the piston pin from the piston assembly.

The piston and the connecting rod can now be separated.

It is recommended that these parts be kept together for possible analysis.



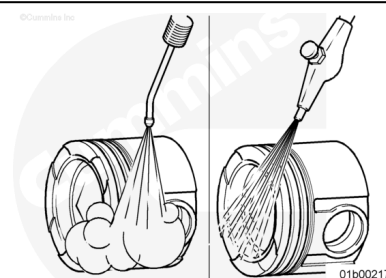
Clean and Inspect for Reuse

Clean and inspect the connecting rod for reuse. Refer to Procedure 001-014 in Section 1.

(/qs3/pubsys2/xml/en/procedures/56/56-001-014.html)

Clean and inspect the pistons for reuse. Refer to Procedure 001-043 in Section 1.

(/qs3/pubsys2/xml/en/procedures/56/56-001-043-tr.html)



Assemble

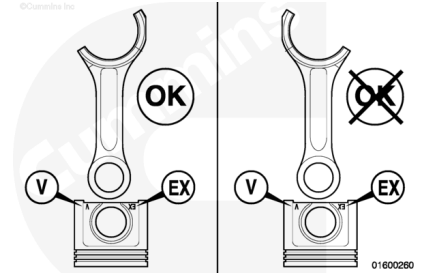
⚠ CAUTION ⚠

To reduce the possibility of severe engine damage, the piston must be oriented correctly on the connecting rod due to the valve pockets in the piston crown.

For correct orientation of the pistons on the connecting rod, there is an “EX” mark on the skirt of the piston.

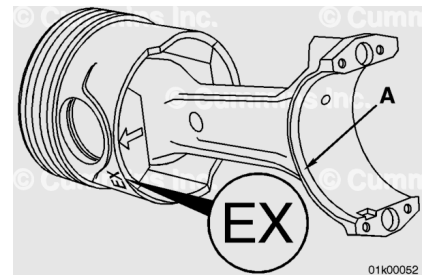
Angle Split Connecting Rod

- The correct orientation is when “EX” mark is by the long side of the connecting rod shank.

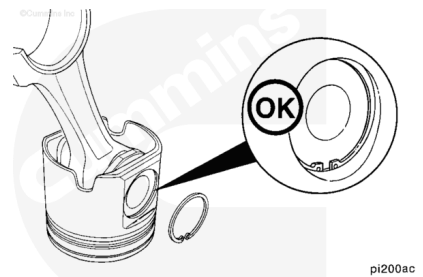


Straight Split Connecting Rod

- The correct orientation is when “EX” mark is by the side of the big chamfer (A) of the connecting rod journal bore.



Install a new snap ring in one piston pin bore of each piston.
It is **not** necessary to heat the pistons before assembly. The piston pin is slip-fit.

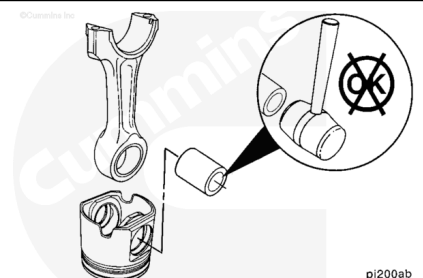


⚠ CAUTION ⚠

Do not use a hammer to install the piston pin. The piston can distort, causing it to seize in the liner.

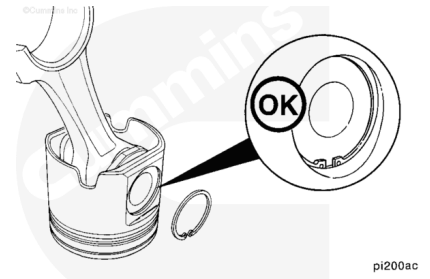
Lubricate the piston pin and connecting rod bushing with clean engine oil.

Align the pin bore of the rod with the pin bore of the piston skirt and crown, and install the piston pin.



The snap ring **must** be seated completely in the piston groove to prevent engine damage during operation.

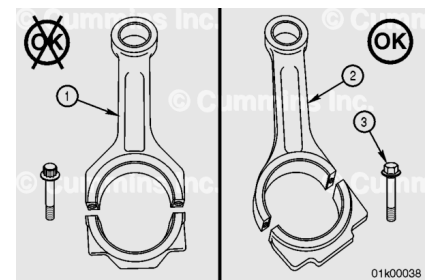
Install a new snap ring in the piston pin bore.



Install

Use the following steps for engines containing angle split connecting rods (2) **only**.

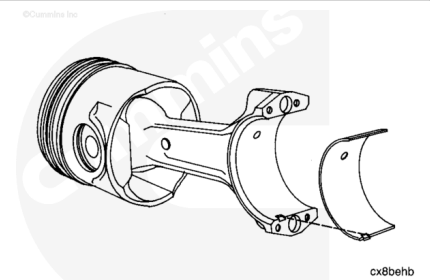
Use the following procedure for engines containing straight split connecting rods (1). Refer to Procedure 001-045 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-045-tr.html>)



Note : The angle split connecting rod (2) can be identified by 6 point head capscrews (3) and the angled joint

Use a lint-free cloth to clean the connecting rod and the bearing shells.

Install the connecting rod bearing. Make sure the tang is positioned as shown. The end of the bearing **must** be even with the cap mounting surface.



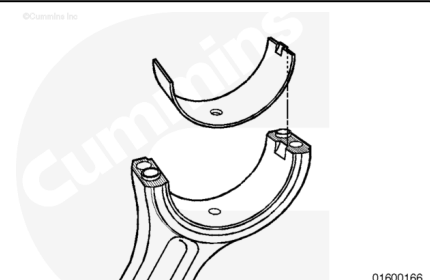
⚠ CAUTION ⚠

To reduce the possibility of engine damage, do not lubricate the backside of the bearing shells.

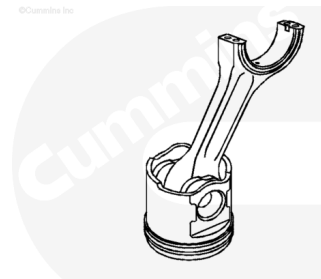
Lubricate the bearing surfaces with Lubriplate™ 105, Part Number 3163086.

The bearings **must** be installed in their original location, if new bearings are **not** used.

Note : All of the rod bearings are identical.



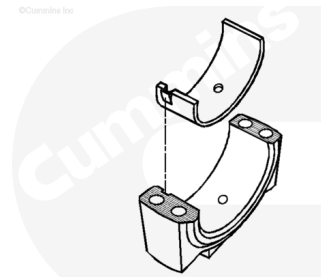
The guides aid the assembly and protect the crankshaft.
Install two connecting rod guides, or equivalent, in the rod.



01600165

⚠ CAUTION ⚠

The connecting rods and caps are not interchangeable. The rods and the caps are machined as an assembly. Failure will result if they are mixed.

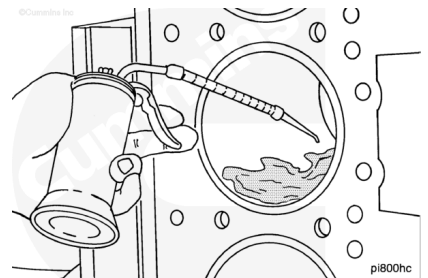


01600167

Install the lower bearing shell in the rod cap. Make sure the tang of the bearing shell is in the slot of the cap and the end of the bearing is even with the cap surface.

Lubricate the bearing surface with Lubriplate™ 105, Part Number 3163086.

Lubricate the cylinder liner with clean engine oil. The entire bore **must** be lubricated.

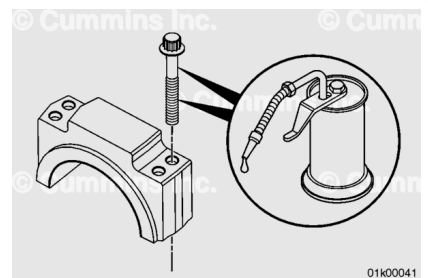


pi800hc

Lubricate the connecting rod capscrews head flanges with clean Society of Automotive Engineers (SAE) W140 gear oil.

Lubricate the capscrew threads with clean engine oil.

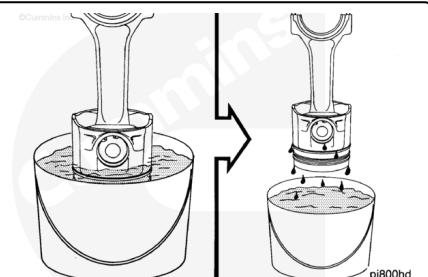
Install the washers and capscrews in the caps.



01K00041

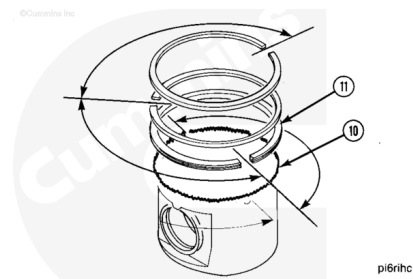
Immerse the piston in engine oil until the rings are covered.

Allow the excess oil to drip off the assembly.



pi800hd

Check to make sure the ring gap position is still correct.

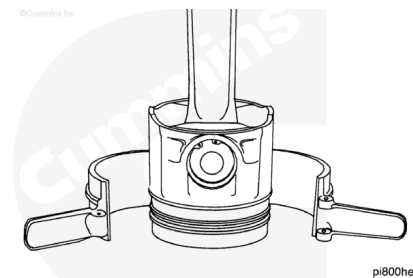


⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the rings fit correctly in the piston.

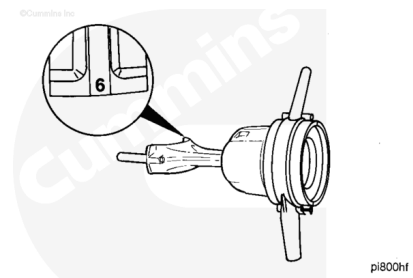
The ring compressor has a tapered bore. The small end of the taper **must** be positioned toward the piston skirt.

Use a piston ring compressor, Part Number 3375342, or equivalent. Install the ring compressor on the piston.



⚠ CAUTION ⚠

The connecting rod and connecting rod cap are a matched set. The cylinder number on the rod and cap must be the same. Damage to the engine can occur if the rods and rod caps are mixed.



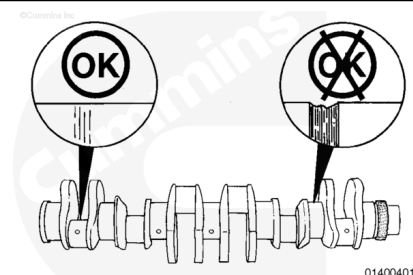
⚠ CAUTION ⚠

The side of the connecting rod with the bearing tang must be on the inboard side of the engine. If the connecting rods are installed backward, engine damage will occur.

Rotate the crankshaft until the journal for the connecting rod being installed is at top dead center and centered in the cylinder bore.

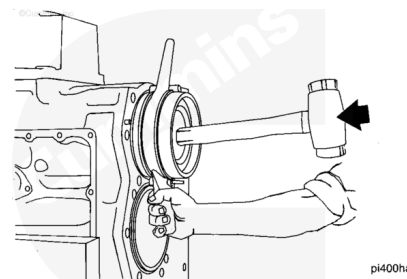
Note : The crankshaft is shown out of the engine for clarity.

Check the crankshaft rod journals for damage.



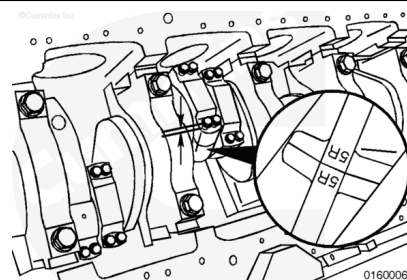
Install the connecting rod and piston until the ring compressor touches the cylinder block. Align the rod with the crankshaft journal.

Hold the ring compressor firmly against the block. Use a wooden hammer handle to push the piston into the liner.



⚠ CAUTION ⚠

The connecting rod and connecting rod cap are a matched set. The cylinder number on the rod and cap must be the same, damage to the engine can occur if the rods and rod caps are mixed.



⚠ CAUTION ⚠

The side of the connecting rod with the bearing tang must be on the inboard side of the engine. If the connecting rods are installed backward, engine damage will occur.

Install the connecting rod cap. Tighten the capscrews alternately and evenly to pull the cap over the dowel pins.

Use the following steps and sequence to tighten the capscrews.

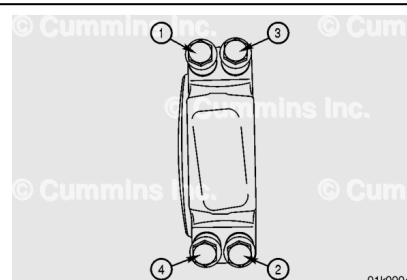
Use an electronic torque wrench to tighten the connecting rods capscrews.

Use sequence 1, 2, 3, 4 on each connecting rod when tightening the connecting rod capscrews. See the graphic.

Torque Value:

1. 85 N•m [63 ft-lb]
2. Repeat step 1
3. Loosen all capscrews 2 revolutions minimum
4. 85 N•m [63 ft-lb]
5. Repeat step 4
6. Advance 60 degrees

If the final torque falls outside the range of 130 to 165 N•m [96 to 122 ft-lb], repeat steps 3 through 6.



Use the following torque procedure if accessing the connecting rod cap through the handhole cover if the torque angle dial is inaccessible.

Prior to applying this process, be sure that all capscrews have been tightened by hand as far as possible to prevent thread damage or mis-engagement.

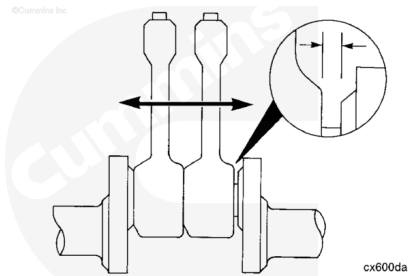
Use the following steps and sequence to tighten the capscrews.

Use sequence 1, 2, 3, 4 on each connecting rod when tightening the connecting rod capscrews. See the graphic.

- Torque Value:**
- 1. 85 n•m [63 ft-lb]
 - 2. Repeat step 1
 - 3. Loosen all capscrews two revolutions minimum
 - 4. 85 n•m [63 ft-lb]
 - 5. Repeat step 4
 - 6. 140 n•m [103 ft-lb]
 - 7. Repeat Step 6

Check the side clearance between the connecting rod and the crankshaft. Refer to Procedure 001-005 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-005-tr.html)

The connecting rod **must** move freely from side to side.



Connecting Rod and Crankshaft Side Clearance

mm		in
0.30	MIN	0.012
0.52	MAX	0.020

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Connect the batteries. See equipment manufacturer service information.
- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-046-tr.html)
- Install the lubricating oil pump. Refer to Procedure 007-031 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html)
- Install the lubricating oil transfer tube. Refer to Procedure 007-040 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-040-tr.html)
- Install the lubricating oil pan adapter. Refer to Procedure 007-027 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/56/56-002-004-tr.html)
- Install new oil filter elements. Refer to Procedure 007-013 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html)
- Fill the engine with clean engine lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)
- Perform the engine run-in procedure. Refer to Procedure 014-002 in Section 14.
(/qs3/pubsys2/xml/en/procedures/56/56-014-002.html)